Multi-Pitch Assistant

Use Case Specifications

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UC-01 – Start Climbing Session	
Primary Actor	Climber
Secondary Actors	None
Description	The climber initiates a new climbing session, enabling the system to begin tracking altitude changes and recording events.
Trigger	The climber taps the 'Start Climbing' button on the home screen.
Preconditions	PRE-01.1 The app is installed and running on the device. PRE-01.2 No other climbing session is currently active.
Postconditions	POST-01.1 GPS location is saved, if GPS permissions are granted. POST-01.2 A new session record is created, with a predefined first event 'session-started', using the saved location. POST-01.3 Session is marked as 'active', ready to begin reading altitude. POST-01.4 Altitude monitoring is running.
Normal Flow	1. The climber opens the app. 2. The system displays the home screen with the 'Start Climbing' button. 3. The climber taps 'Start Climbing'. 4. GPS permission is asked. 5. The system creates a new session record with a unique ID and start timestamp. 6. The system begins monitoring altitude via the device barometer. 7. The system logs a 'session-started' event. 8. The system transitions to the active session screen showing a live altitude reading and an empty event feed.
Alternative Flows	ALT-01.A GPS is unavailable: A.1 The system detects that GPS is not available. A.2 The system proceeds without location data; all other steps continue normally from step 5. ALT-01.B Barometer is unavailable B.1 The system does not detect a barometer sensor. B.2 Automatic event detection is disabled. B.3 The climber may choose to proceed with manual note-taking only, or cancel.
Exceptions	EXC-01.A GPS request throws an unexpected error: A.1 The system catches the error and proceeds without location data. A.2 The session continues normally from step 4 as if GPS was unavailable.

UC-02 – End Climbing Session	
Primary Actor	Climber
Secondary Actors	None
Description	The climber ends an active climbing session and the system persists the complete session record.
Trigger	The climber taps the 'End Climbing' button on the active session screen.
Preconditions	PRE-02.1 A climbing session is currently active.
Postconditions	POST-02.1 A 'session-ended' event has been logged with the current timestamp and current altitude. POST-02.2 Session is marked as inactive. POST-02.3 Altitude monitoring is stopped. POST-02.4 The session is visible in the session history.
Normal Flow	1. The climber taps 'End Climbing' on the active session screen. 2. The climber is prompted for confirm. 3. The system logs an ‘session-ended event’ with the current altitude and timestamp. 4. Session is marked as inactive. 5. Altitude recording is stopped. 6. The system reverts to the home screen. 7. Session history is available in the ‘Log’ tab.
Alternative Flows	ALT-02.A App is closed or crashes while a session is active: A.1 The system detects on next launch that an unfinished session exists. A.2 The system adds an ‘error’ event in the Session. A.3 The system automatically ends the session and saves it with the last recorded timestamp and altitude.
Exceptions	EXC-02.A Session record fails to save: A.1 The system notifies the climber that the session could not be saved. A.2 The climber may retry saving or discard the session.

UC-03 – Monitor Altitude in Real-Time	
Primary Actor	Climber
Secondary Actors	None
Description	While a session is active, the system continuously reads barometric pressure, converts it to altitude, and displays the current altitude to the climber. Altitude deltas are fed to the automatic event classifier.
Trigger	A climbing session becomes active (UC-01 completes successfully).
Preconditions	PRE-03.1 A climbing session is active. PRE-03.2 The device barometer sensor is available and permission has been granted.
Postconditions	POST-03.1 The current altitude is shown in real-time on the active session screen. POST-03.2 Each significant altitude delta has been forwarded to the event classifier.
Normal Flow	1. The system starts a polling loop on the device barometer at a fixed interval. 2. The system reads the current pressure and converts it to an altitude value. 3. The system calculates the delta between the current and previous altitude reading. 4. The system checks whether the delta exceeds the noise-filter threshold. 5. If the delta exceeds the threshold, the system forwards it to the event classifier (see UC-04, UC-05, UC-06). 6. The system updates the displayed altitude on the active session screen. 7. Steps 1–6 repeat until the session ends.
Alternative Flows	ALT-03.A Delta is below the noise-filter threshold: A.1 The system discards the reading. A.2 The displayed altitude is not updated; the loop continues from step 1.
Exceptions	EXC-03.A A barometer reading returns an error: A.1 The system logs an 'error' event. A.2 The system continues the polling loop; monitoring is not interrupted.

UC-04 – Detect Fall	
Primary Actor	Climber
Secondary Actors	None
Description	The system automatically detects a fall by analysing a sudden negative altitude delta and records it as a 'fall' event in the active session.
Trigger	The event classifier receives a negative altitude delta that meets or exceeds the fall-detection threshold.
Preconditions	PRE-04.1 A climbing session is active. PRE-04.2 Altitude monitoring (UC-03) is running.
Postconditions	POST-04.1 A 'fall' event has been logged with timestamp and altitude. POST-04.2 The fall event is visible in the active session event feed.
Normal Flow	1. The event classifier receives a negative altitude delta from UC-03. 2. The system detects if there is upward movement (altitude gain) in the reading. 3. The system evaluates the delta against the fall-detection threshold. 4. The delta meets or exceeds the threshold. 5. The system classifies the movement as a fall. 6. The system logs a 'fall' event with the current timestamp and altitude. 7. The system displays the fall event in the active session event feed.
Alternative Flows	None
Exceptions	EXC-04.A The delta is ambiguous (between fall and retreat thresholds): A.1 The system applies additional checks (e.g. rate of change, duration, controlled descend). A.2 If the movement is classified as a retreat, UC-06 handles logging instead.

UC-05 – Detect Pitch Change	
Primary Actor	Climber
Secondary Actors	None
Description	The system automatically detects when the climber completes a pitch by identifying a significant sustained upward altitude change, and records a 'pitch-changed' event.
Trigger	The event classifier receives a positive altitude delta pattern consistent with completing a pitch.
Preconditions	PRE-05.1 A climbing session is active. PRE-05.2 Altitude monitoring (UC-03) is running.
Postconditions	POST-05.1 A 'pitch-changed' event has been logged with timestamp and altitude. POST-05.2 The ‘pitch-changed’ event is shown on the active session screen.
Normal Flow	1. The event classifier receives a positive altitude delta from UC-03. 2. The system evaluates the altitude gain pattern over a time window. 3. The pattern is consistent with a completed pitch. 4. The system increments the session's pitch count. 5. The system logs a 'pitch-changed' event with the current timestamp and altitude. 6. The system updates the pitch count display on the active session screen.
Alternative Flows	ALT-05.A Upward movement is brief and does not cross the pitch threshold: A.1 The classifier does not fire; no event is logged. A.2 Monitoring continues normally.
Exceptions	None

UC-06 – Detect Rest or Retreat	
Primary Actor	Climber
Secondary Actors	None
Description	The system automatically classifies prolonged low-movement or a sustained downward altitude trend as a rest or retreat respectively, and logs the appropriate event.
Trigger	The event classifier receives altitude deltas that, over a time window, meet the criteria for rest or retreat.
Preconditions	PRE-06.1 A climbing session is active. PRE-06.2 Altitude monitoring (UC-03) is running.
Postconditions	POST-06.1 A 'rest' or 'retreat' event has been logged with timestamp and altitude. POST-06.2 The event is visible in the active session event feed.
Normal Flow	1. The event classifier accumulates altitude deltas over a rolling time window. 2. The system evaluates the overall movement pattern. 3a. If the cumulative altitude change is within the rest threshold for the required duration, the system classifies the period as a rest. 3b. If a sustained downward trend is detected (above the fall threshold), the system classifies it as a retreat. 4. The system logs the appropriate event ('rest' or 'retreat') with timestamp and altitude. 5. The event appears in the active session event feed.
Alternative Flows	ALT-06.A Climber resumes upward movement before rest timeout elapses: A.1 The system resets the rest timer. A.2 No rest event is logged; monitoring continues normally.
Exceptions	None

UC-07 – Add Manual Note	
Primary Actor	Climber
Secondary Actors	None
Description	During an active session the climber manually records a text note, which is saved as a timestamped event in the session.
Trigger	The climber taps the '+' (add note) button on the active session screen.
Preconditions	PRE-07.1 A climbing session is currently active.
Postconditions	POST-07.1 A 'manual-note' event has been logged with the climber's text, timestamp, and current altitude. POST-07.2 The note is visible in the active session event feed.
Normal Flow	1. The climber taps the '+' button on the active session screen. 2. The system displays a text input dialog. 3. The climber enters a note. 4. The climber confirms the note. 5. The system captures the current timestamp and altitude. 6. The system saves a 'manual-note' event containing the text, timestamp, and altitude. 7. The system closes the dialog and the note appears in the event feed.
Alternative Flows	ALT-07.A Climber cancels the dialog: A.1 The system closes the dialog without saving. A.2 The session continues normally with no event logged.
Exceptions	None

UC-08 – View Session History	
Primary Actor	Climber
Secondary Actors	None
Description	The climber browses a chronological list of all completed climbing sessions stored on the device.
Trigger	The climber navigates to the 'Log' tab in the app's navigation.
Preconditions	PRE-08.1 The app is running. PRE-08.2 At least one completed session exists in the local storage.
Postconditions	POST-08.1 A list of completed sessions is displayed, each showing its name, date, climbing type, and pitch count.
Normal Flow	1. The climber taps the 'Log' tab. 2. The system retrieves all completed sessions from local storage, ordered by start date descending. 3. The system displays a scrollable list of session summary cards. 4. Each card shows: session name, date, location, total duration. 5. The climber scrolls through the list.
Alternative Flows	ALT-08.A No completed sessions exist: A.1 The system displays an empty-state message indicating no sessions have been recorded yet.
Exceptions	EXC-08.A Session data cannot be loaded (storage error): A.1 The system displays an error message. A.2 The climber may attempt to reload or return to the home screen.

UC-09 – View Session Details	
Primary Actor	Climber
Secondary Actors	None
Description	The climber selects a session from the history list to view its full data and a complete chronological log of all recorded events.
Trigger	The climber taps a session summary card in the session history list (UC-08).
Preconditions	PRE-09.1 The climber is on the session history screen. PRE-09.2 The selected session record exists in local storage.
Postconditions	POST-09.1 The session detail screen is displayed, showing all data and a full event log.
Normal Flow	1. The climber taps a session card. 2. The system loads the full session record including all associated events. 3. The system displays the session detail screen with: name, start/end date-time, location. 4. Below the metadata, the system lists every event in chronological order. 5. Each event entry shows its type, timestamp, altitude (where available), and any notes. 6. The climber reviews the information. 7. The climber can edit data such as name.
Alternative Flows	None
Exceptions	EXC-09.A The session record fails to load: A.1 The system displays an error message. A.2 The climber is returned to the session history screen.

UC-10 – Edit Event Information	
Primary Actor	Climber
Secondary Actors	None
Description	The climber edits the events of an existing event within a completed session.
Trigger	The climber swipes right and taps the edit button on the session detail screen (UC-09).
Preconditions	PRE-10.1 The climber is viewing the detail screen of a completed session. PRE-10.2 The event is not a 'session-started' or 'session-ended' event.
Postconditions	POST-10.1 The selected event's type, altitude and notes field has been updated. POST-10.2 The updated information is reflected on the session detail screen and in the history list.
Normal Flow	1. The climber swipes right on an event in the event list. 2. The system presents an editable text field pre-populated with the event's existing type, altitude and notes. 3. The climber modifies the type, altitude, notes. 4. The climber confirms the edit. 5. The system saves the updated event notes. The system returns to the session detail screen showing the updated event.
Alternative Flows	ALT-10.A Climber cancels the edit: A.1 The system discards the changes and returns to the session detail screen.
Exceptions	None

UC-11 – Delete Event	
Primary Actor	Climber
Secondary Actors	None
Description	The climber removes an existing event from a session.
Trigger	The climber swipes left and taps the delete button on the session detail screen (UC-09).
Preconditions	PRE-11.1 The climber is viewing the detail screen of a completed session. PRE-11.2 The event is not a 'session-started' or 'session-ended' event.
Postconditions	POST-11.1 The event has been permanently removed from the session record.
Normal Flow	1. The climber swipes left and picks the delete option on an event. 2. The system removes the event from the session record. 3. The system returns to the session detail screen with the event removed.
Alternative Flows	None
Exceptions	None

UC-12 – Add Event to Session	
Primary Actor	Climber
Secondary Actors	None
Description	The climber manually adds a new event to an existing session from the session detail screen.
Trigger	The climber taps the add button at the bottom left of the screen on the session detail screen.
Preconditions	PRE-12.1 The climber is viewing the session detail screen (UC-09).
Postconditions	POST-12.1 A new event has been added to the session record.
Normal Flow	1. The climber taps the add event button. 2. The system presents a form with a mandatory timestamp selector, mandatory event type selector, optional altitude and notes field. 3. The climber selects an event type, inputs the timestamp and optionally enters notes/altitude. 4. The climber confirms. 5. The system creates a new event with the selected type, current timestamp, and notes. 6. The system saves the event to the session record. 7. The session detail screen updates to show the new event in the event list.
Alternative Flows	ALT-12.A Climber cancels: A.1 The system discards the action and returns to the session detail screen unchanged.
Exceptions	None